



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE(Advanced)
REVIEW TEST
15-08-2021

JEE(Main + Advanced) : ENTHUSIAST COURSE (PHASE : TEAS)

Time : 3 Hours

PAPER-2

Maximum Marks : 198

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **24** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

Please see the last page of this booklet for rest of the instructions

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140

- **Boltzmann constant** $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$
- **Coulomb's law constant** $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$
- **Universal gravitational constant** $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$
- **Speed of light in vacuum** $c = 3 \times 10^8 \text{ ms}^{-1}$
- **Stefan-Boltzmann constant** $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$
- **Wien's displacement law constant** $b = 2.89 \times 10^{-3} \text{ m-K}$
- **Permeability of vacuum** $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$
- **Permittivity of vacuum** $\epsilon_0 = \frac{1}{\mu_0 c^2}$
- **Planck constant** $h = 6.63 \times 10^{-34} \text{ J-s}$

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

SECTION-I : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

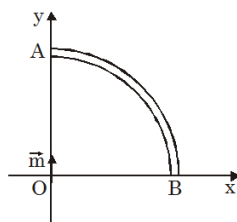
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

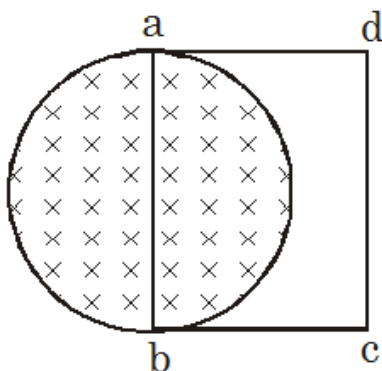
1. Consider a quarter circular conducting ring of large radius $r = 1\text{m}$ with its centre at the origin, where a magnetic dipole of moment 0.4 Am^2 is placed as shown in the figure. When the ring rotates at constant angular velocity 400 rad/s about the y-axis in anticlockwise direction as seen from above.



- (A) emf is induced between its ends such that end B is at a higher potential
 (B) emf is induced between its ends such that end A is at a higher potential
 (C) magnitude of emf between A and B is $16\mu\text{V}$
 (D) magnitude of emf between A and B is $8\mu\text{V}$

Space for Rough Work

2. A uniform magnetic field B exists in a cylindrical region of radius 10 cm as shown in figure. A uniform wire of length 80 cm and resistance 40 ohms is bent into a square frame and is placed with one side along a diameter of cylindrical region. If the magnetic field increases at a constant rate 0.1 tesla per second, find the correct statement(s).



- (A) EMF in ab is zero. (B) Potential difference in ab is zero
 (C) potential difference in ab is $\frac{\pi}{8}$ mV (D) current in the loop is $\frac{\pi}{80}$ mA

Space for Rough Work

3. A point source emits a spherical sound wave at frequency $f = 400\text{Hz}$ with power $P = \pi\text{W}$ in all directions. An observer A is at distance $R = 200\text{ m}$ and does not move, a second observer B is at the same distance and moves with velocity of 60 m/s towards the source. In case C, an identical source moves towards a stationary observer C with a speed of 60 m/s . At the instant of recording, he is at a distance of 200 m from the source. Use speed of sound as 340 m/s .
- (A) A records minimum frequency (B) C records maximum frequency
(C) Intensity recorded by them are $I_A < I_B < I_C$ (D) Intensity recorded by them are $I_C < I_A < I_B$
4. A composite body consists of two rectangular plates of the same dimensions but different thermal conductivities K_A and K_B . This body is used to transfer heat between two objects maintained at different temperatures. The composite body can be placed such that flow of heat takes place either parallel to the interface or perpendicular to it. Calculate the effective thermal conductivities $K_{\parallel}$ and $K_{\perp}$ of the composite body for the parallel and perpendicular orientations. Which of the following option(s) are correct ?
- (A) $K_{\parallel} = \frac{K_A + K_B}{2}$ (B) $K_{\perp} = \frac{K_A + K_B}{2}$
(C) $K_{\parallel} > K_{\perp}$ (D) $K_{\parallel} < K_{\perp}$

Space for Rough Work

5. We want to find the wavelength of a beam of monochromatic light using the fringe width in a YDSE apparatus. $D = 1.00 \pm 0.01$ m, $d = 2.00 \pm 0.01$ mm, $\beta = 0.25 \pm 0.01$ mm. Symbols have their usual meanings. Choose the correct statement(s):
- (A) The estimated wavelength is 500 nm
 - (B) The uncertainty in the wavelength is 5.5%
 - (C) The experiment is most accurate when we measure the fringe width near the central maxima
 - (D) The experiment is most accurate when we measure the fringe width far from the central maxima
6. The radionuclide ^{56}Mn has a half-life of 3 hrs and is produced in a cyclotron by bombarding a manganese target with deuterons. The target contains only the stable manganese isotope ^{55}Mn . If the bombardment lasts much longer than 3 hrs, the activity of ^{56}Mn reaches a steady state value of 6.93×10^{10} decay per sec. Assume that all the incident deuterons cause nuclear reaction.
- (A) Apart from ^{56}Mn , a proton is also produced in the nuclear reaction
 - (B) The deuterons are incident on the sample at a rate of 6.93×10^{10} per sec
 - (C) The steady state mass of ^{56}Mn is approx. 100 ng. (here ng is nano gram)
 - (D) Apart from ^{56}Mn , a neutron is also produced in the nuclear reaction

Space for Rough Work

SECTION-II : (Maximum Marks: 24)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

1. A thin wire of uniform cross-section is drawn from a piece of mass 1 kg of a metal of density 8000 kg/m^3 and resistivity $2 \Omega\text{m}$. Now the ends of the wire are joined to make a planar closed loop. The loop is placed in a uniform magnetic field whose induction varies with time t according to law $B = \pi t$ tesla. What is the maximum current (in μA) you can obtain in the loop by properly selecting shape and dimensions of the loop ?
2. A certain length $\ell = 90 \text{ cm}$ of a sonometer wire resonates with a tuning fork when loaded with a mass of density 9 gm/cc . When the mass is immersed in liquid A the length of the wire resonating with the same tuning fork is 70 cm , when it is immersed in a liquid B the length is 50 cm . Calculate the density of the liquid B/ density of liquid A. In all the cases, the string is oscillating in the fundamental mode.
3. A U tube of height 1 m on both the sides is filled with mercury upto a height of 50 cm . The air above is diatomic and is at atmospheric pressure of 100 kPa . It always remains at room temperature. Mercury column of length 50 cm is now poured in the right limb and then the tube is sealed at upper end of left limb. The air column on the left is made to resonate with a sound source fitted inside the air column. What is the fundamental frequency (in Hz) of the air column? It is given that at room temperature and atmospheric pressure, the velocity of sound in air is 320 m/s .

Space for Rough Work

4. One mole of monoatomic gas undergoes a polytropic process such that $V = CT^3$. The symbols have their usual notations. What is the heat absorbed (in J) by the gas if its temperature increases by 2 K ? [Take $R = 25/3$ J/mol K.]
5. In a cube, the edge is measured using vernier calliper scale. In this calliper, 9 divisions of the main scale is equal to 10 divisions of vernier scale and 1 main scale division is 1 mm. The main scale division reading is 10 and 1 division of vernier scale was found to be coinciding with the main scale. The mass of the cube is 2.736 g. Calculate the density in gm/cc.
6. An electron and a proton are separated by a large distance. The electron starts approaching the proton with energy 2 eV. The proton captures the electron and forms a hydrogen atom in first excited state. The remaining energy of electron is emitted in the form of a photon. The resulting photon is incident on a photosensitive metal of threshold wavelength $4600\text{\AA}$. What is the maximum kinetic energy of the emitted photoelectron (in eV)?

Space for Rough Work

SECTION-III : (Maximum Marks: 18)

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Full Marks : +3 If only the correct answer is given.

Zero Marks : 0 If no answer is given.

Negative Marks : -1 In all other cases

1. A chamber at small height h above earth's surface maintained at 30°C has a clock fitted in it with an uncompensated pendulum. The clock designer correctly designs it for height h , but for temperature of 20°C . If this chamber is taken to earth's surface where temperature is also 30° , the clock in it would click correct time. Find the coefficient of linear expansion of material of pendulum. (earth's radius is R) If the coefficient is h/nR , find the value of n .
2. The temperature of ice is -20°C (specific heat of ice = $0.5 \text{ cal/g}^\circ\text{C}$) and that of water 60°C . They are mixed in equal amounts of 8 kg each. What mass of ice (in kg) will be melted? (Latent heat of fusion of ice = 80 kcal/kg)
3. The number of significant digits in the answer of $\sqrt{(2.8 - 2.71)}$ is
4. Two identical sources of sound S_1 and S_2 produce intensity I_0 at a point P equidistant from each source. If the power of S_1 is reduced by 64%, then determine the maximum and minimum intensities at the point P.
The value of $\sqrt{\frac{I_{\max}}{I_{\min}}}$ is
5. If fast moving electrons are used in a YDSE apparatus, the fringe width comes out to be 100 nm. What is the fringe width (in μm) when we use photons of the same energy as that of the fast moving electrons? Take $D = 2.6 \text{ m}$, $d = 2.21 \text{ mm}$, $h = 6.63 \times 10^{-34} \text{ Js}$.
6. Find the coulombic force of repulsion (in N) between a proton at the surface of a ${}^{64}_{29}\text{Cu}$ nucleus and the remaining 28 protons. Assume that remaining nucleus is spherically symmetric and $R_0 = 1.2 \text{ fm}$. Fill $\frac{F}{40}$ in the OMR sheet

Space for Rough Work

PART-2 : CHEMISTRY
SECTION-I : (Maximum Marks: 24)

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1. Which of the following changes can be experimentally realized in the laboratory:

- (A) Combustion of Mg ribbon in CO_2
(B) Reduction of FeCl_3 (aq.) on adding Zn granules
(C) Direct heating of hydrated MgCl_2 to get anhydrous MgCl_2
(D) Evolution of brown gas on heating LiNO_3

2. Colour of iodine solution is disappeared by shaking it with aqueous solution of :-

- (A) $\text{S}_2\text{O}_3^{2-}$ (B) NO_2^- (C) CO_3^{2-} (D) Cl_2

Space for Rough Work

3. Choose correct statement :-

- (A) In the balanced redox reaction $x\text{As}_2\text{S}_3 + y\text{NO}_3^- + 4\text{H}_2\text{O} \rightarrow a\text{AsO}_4^{3-} + b\text{NO} + c\text{SO}_4^{2-} + 8\text{H}^+$ molar ratio of $x : y = 3 : 28$
- (B) When NH_4SCN oxidizes into SO_4^{2-} , CO_3^{2-} and NO_3^- , its equivalent weight will be $M/24$
- (C) When Bi_2S_3 converted into Bi^{5+} and S , n-factor will be 7
- (D) Equivalent weight of H_3PO_2 when it disproportionate into PH_3 and H_3PO_3 is $3M/4$

4. Which of the following are correct for real gases

- (A) Value of 'a' is measure of magnitude of intermolecular attractive forces within the gas and is independent of temperature and pressure
- (B) Repulsive interactions between molecules are short range interactions and are significant when molecules are almost in contact
- (C) Boyle point of a real gas depends upon nature of gas
- (D) Real gases shows ideal behavior when conditions of temperature and pressure are such that the intermolecular forces are practically negligible.

Space for Rough Work

5. How many of the following will produce fruity odour with acetic acid when heated with Con. H_2SO_4 ?
- (A) $\text{CH}_3 - \text{CH}_2 - \text{OH}$ (B) $\text{CH}_3 - \text{CH}_2 - \text{Cl}$
- (C) $\text{Ph} - \text{Br}$ (D) 
6. Which of the following can be used to convert Ethanal into Ethanol?
- (A) LiAlH_4 (B) $\text{H}^+ / \text{K}_2\text{Cr}_2\text{O}_7$
- (C) NaBH_4 (D) $\text{B}_2\text{H}_6 / \text{H}_2\text{O}_2 / \text{OH}^-$

Space for Rough Work

SECTION-II : (Maximum Marks: 24)

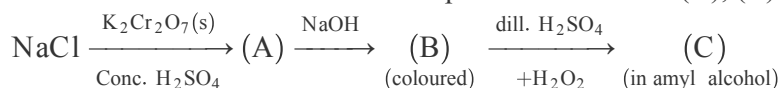
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- Answer to each question will be evaluated according to the following marking scheme:

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1. Calculate the difference in net stabilisation energies (in kJ mol^{-1}) of Fe^{2+} complexes with CN^- ligands ($\Delta_0 = 25000 \text{ cm}^{-1}$) and with H_2O ligands ($\Delta_0 = 10000 \text{ cm}^{-1}$). Given that the pairing energy, $P = 15000 \text{ cm}^{-1}$ for Fe^{2+} ion for both the complexes.
(Given that $h = 6 \times 10^{-34} \text{ J sec.}$, Avogadro Number = 6×10^{23} , Speed of light in vacuum = $3 \times 10^8 \text{ m sec}^{-1}$).

2. Determine total number of atoms in per unit formula of (A), (B) & (C)

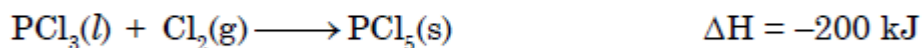
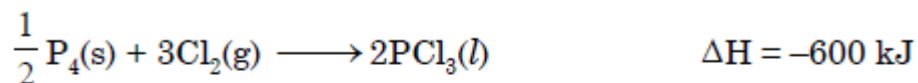


3.

	(List-I) Orbital		(List-II)- Angular dependence function $A(\theta, \phi)$
P	p_z	1	$\frac{\sqrt{15}}{2\sqrt{\pi}} \cos \theta \sin \theta \cos \phi$
Q	d_{xy}	2	$\frac{\sqrt{3}}{2\sqrt{\pi}} \cos \theta$
R	d_{zx}	3	$\frac{\sqrt{15}}{2\sqrt{\pi}} \sin^2 \theta \sin \phi \cos \phi$

Space for Rough Work

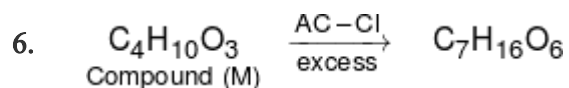
4. Given that



Heat of formation of $\text{PCl}_5(\text{s})$ is $(-x \times 10^2 \text{ kJ})$, value of x is :

5. How many of the following will produce methane gas when treated with CH_3MgBr ?

- (a) Acetic acid (b) carbolic acid (c) Benzaldehyde (d) Benzonitrile
 (e) Wood spirit (f) Salicylic acid (g) Hexachlorobenzene
 (h) But-2-yne (i) Acetylene



How many $-\text{OH}$ groups are present in compound M?

Space for Rough Work

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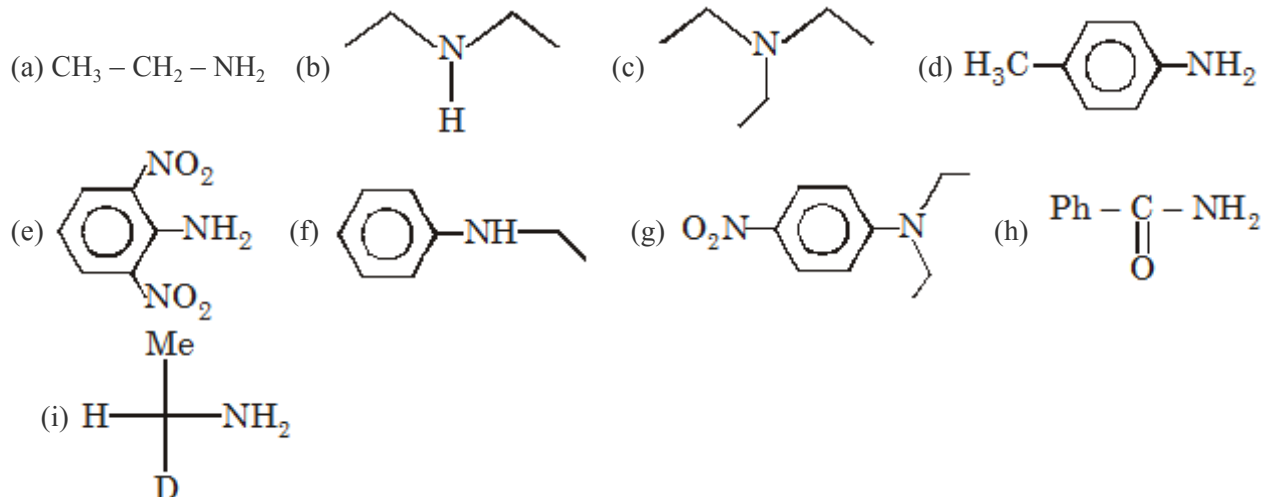
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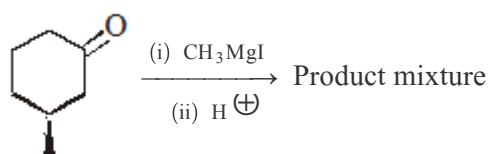
1. In salt analysis among the following, total number of acidic radicals gives colourless volatile gas/vapour as a observation with dilute H_2SO_4 is.
 $\text{C}_2\text{O}_4^{2-}$, Cl^- , Br^- , NO_2^- , NO_3^- , CH_3COO^- , CO_3^{2-} , SO_3^{2-}
2. $\text{ScF}_3 + 3\text{NH}_4\text{F} \rightarrow 3\text{NH}_4^+ + [\text{A}]$ complex ion
 Find the sum of nodal plane of d orbitals involved in hybridisation of [A].
3. One mole of an diatomic ideal gas in an insulated cylinder fitted with a massless, frictionless movable piston is expanded gradually from 1 L to 32 L. If the initial temperature of gas is 1000 K then find magnitude of change in Gibb's Free energy in kcal.
 (Molar entropy of gas = $3 \text{ cal mol}^{-1}/\text{K}^{-1}$)
 (Use, $R = 2 \text{ cal mol}^{-1} \text{ K}^{-1}$)
 (Assume molar entropy of a gas is constant)
4. How many minimum moles of MgCl_2 should be added to just precipitate $\text{Mg}(\text{OH})_2$ in 100 ml buffer solution containing 0.2M NH_4OH and 0.1 M $(\text{NH}_4)_2\text{SO}_4$?
 $K_b(\text{NH}_4\text{OH}) = 10^{-5}$; $K_{sp}[\text{Mg}(\text{OH})_2] = 10^{-11}$
First multiply your answer with 100. Then fill your answer as sum of digits (excluding decimal places) till you get the single digit answer.

Space for Rough Work

5. How many of the following will give carbylamine test positive?



6. Select number of correct statements about the following reaction



- (A) Product mixture is optically active
- (B) Reaction follows nucleophilic substitution mechanism
- (C) At least one of the product will show geometrical isomerism
- (D) At least one of the product will show optical isomerism
- (E) Reaction follows nucleophilic addition mechanism
- (F) Major organic products are enantiomers of each other
- (G) Major organic products are diastereoisomers of each other

Space for Rough Work

PART-3 : MATHEMATICS
SECTION-I : (Maximum Marks: 24)

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1. Let $(1 + \sqrt{2})^n = A_n + B_n \sqrt{2}$, where A_n and B_n are rational numbers and $n \in \mathbb{N}$, then the value of $\lim_{n \rightarrow \infty} \frac{A_n}{B_n}$ is equal to -

(A) $\left[\lim_{x \rightarrow 0} \frac{\sin x}{x} \right]$

(B) $\lim_{x \rightarrow 0} \left[\frac{\sin x}{x} \right]$

(C) $\lim_{x \rightarrow \frac{\pi}{4}} \frac{|\cos x - \sin x|}{|x - \frac{\pi}{4}|}$

(D) Does not exist

2. If A is symmetric & B is skew symmetric matrix, then which of the following is/are **INCORRECT**-

(A) ABA^T is symmetric matrix

(B) $AB^T + BA^T$ is symmetric matrix

(C) $(A + B)(A - B)$ is skew symmetric

(D) $(A + I)(B - I)$ is skew symmetric

Space for Rough Work

3. Area enclosed by parabola $ay = 3(a^2 - x^2)$ and x-axis is 64, then value of 'a' can be -
 (A) 4 (B) 2 (C) -4 (D) -2
4. If B is adjoint of matrix A and $B^T B^{-1} = A$ then (where B is non singular matrix)
 (A) B is symmetric matrix (B) $\det(B) = 1$
 (C) A is skew symmetric matrix (D) $\text{adj } B = A$
5. $\vec{a}$, $\vec{b}$ and $\vec{c}$ are three non-zero & non-coplanar vectors with fixed magnitudes. Angles between any of the vector with the normal of the plane containing other two is α , ($\alpha \in (0, \pi)$). Volume of parallelepiped formed is V and its surface area is S. Which of the following is/are true ?
 (A) $\frac{V}{S} = \frac{|\cos \alpha|}{2} \left[\frac{1}{|\vec{a}|} + \frac{1}{|\vec{b}|} + \frac{1}{|\vec{c}|} \right]^{-1}$
 (B) $V < |\vec{a}| |\vec{b}| |\vec{c}|$
 (C) $\vec{a}$, $\vec{b}$ and $\vec{c}$ are equally inclined to each other
 (D) if angle between $\vec{a}$, $\vec{b}$ is θ , then angle between $\vec{c}$ and plane of $\vec{a}$ and $\vec{b}$ is $(90^\circ - \theta)$
6. If the curves $y = 2(x - a)^2$ and $y = e^{2x}$ touches each other, then 'a' is less than-
 (A) -1 (B) 0 (C) 1 (D) 2

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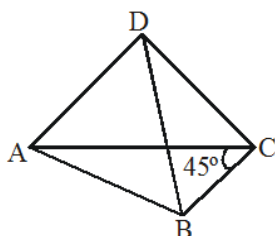
SECTION-II : (Maximum Marks: 24)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

1. As shown in diagram, the volume of tetrahedron ABCD is $\frac{1}{6}$ cu. units. Also $\angle ACB = 45^\circ$, $BC = 1$, $AD + BC + \frac{AC}{\sqrt{2}} = 3$, then length of CD is equal to (in units)



2. Let $L_1 : \vec{r} = (\hat{i} + \hat{j} + \hat{k}) + \lambda(\hat{i} + \hat{j})$, $\lambda \in \mathbb{R}$ and $L_2 : \vec{r} = \hat{i} + (\mu + 1)\hat{j} + (\mu + 1)\hat{k}$, $\mu \in \mathbb{R}$ be the two lines intersecting at point A. Through point B(3, 3, 1) a line is drawn making an angle of 60° with L_2 and intersecting it at point C, then area of ΔABC is equal to

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3. If $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 2015 & 2016 & -1 \end{bmatrix}$, then $(2A^2 + 4A^4 + 6A^6) = \lambda I_3$, then $\frac{\lambda}{10}$ is (I_3 is unit matrix of order 3)-
4. If the value of integral $\int_0^{\pi/4} \frac{\sin^2 7\theta}{\sin 2\theta} d\theta = \frac{2}{3} + k \ln 2$, then the value of k is equal to
5. If 'd' be the degree & 'O' be the order of differential equation of the family of curves $\alpha y^2 = 2x + \alpha$, where α is an arbitrary constant, then (d + O) is
6. If $I_1 = \int_0^{\pi} x \sin x \cdot e^{\cos 4x} dx$ and $I_2 = \int_0^{\pi/2} \cos x \cdot e^{\cos 4x} dx$, then the value of $\left[\frac{I_1}{I_2} \right]$ is
(where $[.]$ denotes greatest integer function)

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SECTION-III : (Maximum Marks: 18)

- This section contains **SIX (06)** questions.
- The answer to each question is a **SINGLE DIGIT INTEGER** ranging from 0 to 9, both inclusive.
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If only the correct answer is given.

Zero Marks : 0 If no answer is given.

Negative Marks : -1 In all other cases

1. If 'e' denotes the eccentricity of the hyperbola, satisfying the differential equation $2xy \frac{dx}{dy} = x^2 + y^2$ and passing through the point (2, 1), then $(e^2 - 1)$ is equal to
2. If $f(x) = (\sin x)^{1/3} \cdot (\sin 2x)^{1/2} \cdot \sin 3x$, then value of $f' \left(\frac{\pi}{3} \right)$ is of the form $-\frac{3^a}{2^b}$, then value of $\left(a - \frac{b}{2} \right)$ is
3. The value of $4 \left(\lim_{x \rightarrow 0} \frac{\frac{\tan^{-1} x}{\int_0^x t \tan^2 t \, dt}}{\frac{\sec x}{\int_1^x (t^2 - 1) \, dt}} \right)$ is

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4. If $L = \lim_{n \rightarrow \infty} \left(\frac{n+2}{n^2+n+1} + \frac{n+4}{n^2+2n+4} + \frac{n+6}{n^2+3n+9} + \dots + \frac{5}{7n} \right)$, then e^L is equal to -
5. If A is a square matrix of order 2 then $\frac{1}{|A|}(-\text{tr}(A^2) + (\text{tr}(A))^2)$ is (Where $|A| \neq 0$)
6. Let $A = [a_{ij}]_{2 \times 2}$ such that $a_{ij} \in \{-1, 0, 1\} \forall i, j$ and $\text{adj}(A) = -A$, $\det(A) = -1$. Then half of the number of such matrices A, is
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NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions
and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form
number of the candidate, and that question
paper and ORS codes are the same.

Signature of the Invigilator

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